

Answer key

Form P · Practice

Marking guide and worked solutions · 40 points · may be shared with students after the review

How to mark. Sections 1 and 5 and the explanation parts of the others reward **sensible reasoning that is defended with a reason or a number**; there is no single right model. Apply **follow-through** from earlier answers. Accept rounding differences of ± 0.02 in s and t and ± 0.5 in forecasts. Students interpret coefficients but are **not yet** asked whether coefficients are significant. In Section 4 a t-test is **significant if $p < 0.05$** (two-tailed, T.TEST type 2). Objective column: syllabus learning outcome (LO) and your category, **Con** concept, **Cal** calculation, **Com** computation in Sheets.

1 · Quick answer sheet

Q	Objective	Expected answer	Pts	How the marks split
1a	LO1, LO5 · Con	Any three sensible inputs with a direction and a reason: weekend (up: more visitors), beach event (up), planned price (down), temperature (either direction with a reason: warm-weather customers may buy cold items instead; the data show down), wind (down: fewer outdoor customers).	3	1 per input: direction and a coherent reason. Either direction for temperature scores if defended.
1b	LO5 · Con	Item 7 , gelato sales counted at closing time: it is not known before opening and it is driven by the same weather and day type, so it cannot be used to forecast.	2	1 identifies item 7 + 1 the problem.
1c	LO3, LO5, LO6 · Con	Not both. $r = 1.00$: they are the same information (the price changed once, on the day the flag switches), so LINEST cannot separate them. Keep the price (the thing the café controls) or the flag, not both.	2	1 identical/overlapping information and why it is a problem + 1 a choice.
1d	LO6 · Con	Hold back some days, fit the model on the rest, predict the held-back days, then calculate their residuals and MAE.	1	1: a hold-out check with residuals/MAE (names the idea of unseen days).
2a	LO5, LO6 · Com	Sales = $84.76 - 0.81 \times \text{temperature} - 0.18 \times \text{wind} + 11.53 \times \text{weekend}$ ($\hat{y} = b + m_1x_1 + m_2x_2 + m_3x_3$; the table shows Col A = m_3 , Col B = m_2 , Col C = m_1 , Col D = b).	2	1 correct numbers matched to the right variables + 1 correct equation form.
2b	LO6, LO7 · Con	(i) Each extra °C goes with about 0.81 fewer pastries a day, holding wind and weekend fixed; pastries still sell on warm days (predictions stay positive). (ii) -0.18 is per km/h: a 10 km/h stronger wind goes with about 1.8 fewer pastries: small, but not “no effect”. (iii) About 33% of the variation in daily sales is explained; R^2 does not say how often the model is wrong; a typical error is about 9 pastries.	3	1 each: corrected statement with units and conditions.
2c	LO3, LO6 · Cal + Con	(a) $5 \times (-0.81) = -4.05$, about -4 pastries . (b) $10 \times (-0.18) = -1.8$ pastries. (c) Possible reasons: customers switch to cold items (gelato, iced coffee) in the heat; temperature stands in for something else that changed over the summer.	3	1 + 1 + 1 (any sensible real-world reason).
2d	LO5, LO6 · Cal	(a) $\hat{y} = 84.76 - 0.81(20) - 0.18(12) + 11.53(1) = 77.9$ pastries. (b) Residual = $98 - 77.9 = +20.1$. (c) $ \text{residual} = 20.1$ is larger than $SE - Y = 8.95$: an unusually large miss (more than $2 \times SE - Y$): the café sold far more than predicted; something not in the model happened (a festival tie-in, a promotion).	4	2 prediction (1 substitution, 1 value) + 1 residual (actual – predicted) + 1 comparison with $SE - Y$.
3a-i	LO3, LO6 · Cal	Filled table: Mon actual 62, predicted 66, residual -4; Sat actual 84, predicted 74, residual +10; Wed actual 58, predicted 60, residual -2; Fri actual 65, predicted 69, residual -4.	1	1 all blanks correct (follow-through on a single slip).
3a-ii	LO3, LO6 · Cal	$MAE = (4 + 10 + 2 + 4) \div 4 = 5$ pastries.	1	1 correct (absolute values used).
3a-iii	LO6 · Con	Model Z has the lower MAE (3 vs 5). But every residual has the same sign: Z over-predicts every day by 3 (systematic bias). A simple correction (subtract 3) would remove it. Model X’s errors are larger but not one-sided.	2	1 lower MAE with numbers + 1 identical signs = bias.

3b	LO3, LO4, LO6 · Con	(i) Day 4 : sales far above the others at a middling temperature. (ii) The slope barely changes (-1.14 with it, -0.98 without) but R^2 jumps from 0.26 to 0.96 ($SE - Y$ 10.3 to 1.2): the spike spoils the fit and the intercept, not the trend. (iii) Find out why (a festival tie-in, a promotion, a recording error). Delete only with a documented reason; otherwise keep it, flag it and report both.	4	1 day + 2 impact (slope barely changes; R^2 changes a lot) + 1 sensible next step.
4a	LO3 · Cal	Means 66 and 63 ; $\Sigma(x - \bar{x})^2 = 40$ and 40 ; $s_1 = \sqrt{(40 \div 4)} = 3.16$, $s_2 = \sqrt{(40 \div 4)} = 3.16$ (deviations in the worked table below).	3	1 means + 1 sums of squares + 1 s values.
4b	LO3 · Cal	$s_p = \sqrt{[(4 \times 10 + 4 \times 10) \div 8]} = \sqrt{10} = 3.16$; $t = (66 - 63) \div [3.16 \times \sqrt{(1/5 + 1/5)}] = 3 \div 2.00 = \mathbf{1.50}$.	2	1 s_p + 1 t (follow-through on s; sign may be reported either way if direction is stated).
4c	LO1, LO3, LO6 · Con	(i) H_0 : the price rise made no difference to mean daily pastry sales. (ii) $p = 0.172 > 0.05$: not significant : no clear evidence of a difference. (iii) No. “Not significant” means the data do not show a clear difference; it does not show there is none (absence of evidence is not evidence of absence). With only $5 + 5$ days a real drop of a few pastries could be missed.	3	1 null hypothesis + 1 decision using p + 1 absence of evidence $\neq$ evidence of absence.
4d	LO6 · Con	More days; or put price with temperature and weekend in a LINEST model; or compare days with similar weather and weekday.	1	1 a relevant reason/idea.
5	LO7 · Con	(i) The inputs recommended (e.g. weekend up, temperature and wind down (the overlapping price and flag cannot both go in)) with the typical forecast error (about 9 pastries). (ii) The price test: pastry sales were a little lower after the price rise but the difference is not significant (about 3 pastries, $p = 0.172$). (iii) A caution (small samples, other changes, association not proof, forecasts can still miss by the typical error).	3	1 inputs + main LINEST finding in plain words + 1 price test result in plain words + 1 a genuine caution.

2 · Worked tables

Section 4 worksheet (Table 6).

Day	Before (x_1)	$x_1 - \bar{x}_1$	$(x_1 - \bar{x}_1)^2$	After (x_2)	$x_2 - \bar{x}_2$	$(x_2 - \bar{x}_2)^2$
1	66	0	0	63	0	0
2	70	+4	16	67	+4	16
3	62	-4	16	59	-4	16
4	68	+2	4	65	+2	4
5	64	-2	4	61	-2	4
Sum	330	0	40	315	0	40
Mean $\bar{x}$	66			63		
s	3.16			3.16		

Check against Sheets: =T.TEST gives $p = 0.1720$ ($df = 8$, $|t| = 1.500$, two-tailed).

3 · Where students go wrong

- **Reading the LINEST table backwards** (2a): Col A is the **last** X column and Col D the intercept. The layout table is printed on the paper, so give no credit for guessing the order; apply follow-through if the order is wrong but the method is right.
- **“Per customer” and “causes”** (2b): the coefficients describe the **daily total** and an **association**.
- **Reading R^2 as “percent correct”** (2b): R^2 is the share of variation explained.
- **Using absolute values** (3a-ii): MAE averages the sizes of the errors, so signs must not cancel.
- **Reading p as the probability that H_0 is true** (4c). Credit “if there were no real effect, a result this extreme would be rare”.